



Academy of
Engineering

(An Autonomous Institute affiliated to Savitribai Phule Pune University)

Kruskal Algorithm

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Code:

```
#include <iostream>
```

```
using namespace std;
```

```
#define MAX 100
```

```
class Edge { public:
```

```
    int u, v, w;
```

```
};
```

```
class Graph {
```

```
    int n, e;
```

```
    Edge edges[MAX];
```

```
    int parent[MAX];
```

```
    public:
```

```

void input() {      cout << "Enter number of
vertices: ";      cin >> n;      cout << "Enter
number of edges: ";      cin >> e;      cout <<
"Enter edges (u v weight):\n";      for (int i = 0; i <
e; i++) {          cin >> edges[i].u >> edges[i].v >>
edges[i].w;
    }

```

```

    // Initialize parent
    for (int i = 0; i < n; i++)
        parent[i] = i;
}

```

```

int find(int i) {
while (parent[i] != i)
i = parent[i];      return
i;
}

```

```

void unite(int i, int j) {
int a = find(i);    int b =
find(j);    parent[a] = b;
}

```

```

void sortEdges() {      for (int i = 0; i
< e - 1; i++) {          for (int j = 0; j < e -
i - 1; j++) {              if (edges[j].w >
edges[j + 1].w) {                  Edge temp

```

```

    = edges[j];          edges[j] = edges[j]
+ 1];          edges[j + 1] = temp;
        }
    }
}
}
}

```

```

void kruskal() {    sortEdges();
int mincost = 0;    cout <<
"\nEdges in MST:\n";
    for (int i = 0; i < e; i++) {        int u = edges[i].u;        int
v = edges[i].v;        if (find(u) != find(v)) {            cout << u
<< " -> " << v << " = " << edges[i].w << endl;            mincost
+= edges[i].w;            unite(u, v);
        }
    }
    cout << "Minimum cost = " << mincost << endl;
}
};

```

```

int main() {
    Graph g;
    g.input();
    g.kruskal();
    return 0; }

```

Output:

```
Enter number of vertices: 4
Enter number of edges: 5
Enter edges (u v weight):
0 1 10
0 2 6
0 3 5
1 3 15
2 3 4
```

Edges in MST:

```
2 -> 3 = 4
0 -> 3 = 5
0 -> 1 = 10
Minimum cost = 19
```

```
Enter number of vertices: 5
Enter number of edges: 7
Enter edges (u v weight):
0 1 4
0 2 2
1 2 1
1 3 5
2 3 8
2 4 10
3 4 2
```

Edges in MST:

```
1 -> 2 = 1
0 -> 2 = 2
3 -> 4 = 2
1 -> 3 = 5
Minimum cost = 10
```